



Practical – 8

Roll No. : S025	Name: Abhishek Pal
Class : S.Y.I.T.	Date/Time of Submission : 11-08-2026

Practical No. 8 - Write Algorithm, Code and Output

Q. 1. Write algorithm for the following:

a. Write a program to accept and pass radius to a function that returns area and circumference (using tuple).

CODE :

```
import math
#Function to calculate area and circumference
def circle_details(r):
    area=math.pi*r*r
    circum=2*math.pi*r
    return (area, circum) #Returning as a tuple
#Accept radius from the user
radius=float(input("Enter radius: "))
#Call Function
result=circle_details(radius)

#Display results
print("Area: ", result[0])
print("Circumference: ", result[1])
print("Abhishek S025")
```

OUTPUT :

```
Enter radius: 5
Area: 78.53981633974483
Circumference: 31.41592653589793
Abhishek S025
```

b. Write a program to perform basic file operations on text files and binary files.

CODE :

```
with open("example.txt", "w") as f:
    f.write("Hello Students ! \n Welcome to Python File Handling")

with open("example.txt", "r") as f:
    content=f.read()

print("Text File contents:\n", content)
print("Abhishek S025")
```

OUTPUT :

```
Text File contents:
Hello Students !
Welcome to Python File Handling
Abhishek S025
```

CODE :

```
with open("b_sample.bin", "wb") as bf:
    data=b"This is binary data"
    bf.write(data)

with open("b_sample.bin", "rb") as bf:
    b_content=bf.read()
    print("Binary File contents:\n", b_content)
print("Abhishek S025")
```

OUTPUT :

```
Binary File contents:
b'This is binary data'
Abhishek S025
```

c. Write a Python program to read last n lines of a file.

CODE :

```
def read_last_n_lines(filename,n):  
    with open(filename, "r") as f:  
        lines=f.readlines()  
        return lines[-n:]  
filename="example.txt"  
n=int(input("Enter number of last line to read:"))  
  
last_lines=read_last_n_lines(filename,n)  
print("\nLast ", n, "Lines : ")  
for line in last_lines:  
    print(line.strip())  
print("Abhishek S025")
```

OUTPUT :

```
Enter number of last line to read:1  
  
Last  1 Lines :  
Welcome to Python File Handling  
Abhishek S025
```